higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING ADMINISTRATION N5

15 APRIL 2019

This marking guideline consists of 6 pages.

Copyright reserved

QUESTION 1

1.1
- Petrol or diesel
- Compressed air
- Electricity (3 × 1) (3)

1.2
1.2.1 Use a light crane or hoists or a rail-mounted crane

1.2.2 Build covered walkways

1.2.3 Let guard with flags control the one entry and exit

1.2.4 Get permission from the authorities to close one lane

1.2.5 Use gantries for storage or store somewhere else

(5 × 1) (5)

1.3
1.3.1 Excavator

1.3.2 Concrete vibrator

1.3.3 Plate vibrator

1.3.4 Pneumatic roller

1.3.5 Scraper

1.3.6 Scotch derrick

1.3.7 Trenching machine

1.3.8 Grader

1.3.9 Mobile crane

1.3.10 Sheep-foot roller

(10 × 1) (10)
[18]

QUESTION 2

2.1
- Review objectives
- Determine major steps
- Establish priorities
- Schedule
- Determine detail
- Review and reconcile (6)

2.2
- Available capital
- How much is liquid and how much is fixed assets
- How much credit or what overdraft facilities are available
- How many times the capital will be turned over
- What is the profit on turnover (Any 3 × 1) (3)

2.3
- Pre-tender planning
- Contract planning
- Detailed or stage planning (3)

2.4 All relevant information for the project is prepared in a systematic manner. (1)
EXAMPLES: It should include access to the site, existing services, labour availability, material availability, type of soil and site requirements. (Any 2 × 1) (2)

2.5 Compare the work completed against the programme. (1)

2.6
- The client may change his/her requirements.
- The design for a certain section may change.
- The weather may have an influence.
- Certain key material may not be available.
- Labour problems may be encountered.
- Problems may become evident during construction. (Any 4 × 1) (4)

2.7 2 850 ÷ 20 = 125 hrs (1)
125 ÷ 8 ÷ 6 = 2,97 weeks or 3 weeks (1)
[22]

QUESTION 3

3.1
- Make available all necessary drawings.
- Give the schedule as per the programme chart.
- Provide space for storage for the subcontractor on site.
- Provide free supply of water.
- Provision for electricity or compressed air should be made.
- Payments should be made as per agreement. (Any 4 × 1) (4)

3.2 The depth of a pile is not known until everything is used. (1)
[5]

QUESTION 4

4.1
- Order and purchase in good time.
- Order correct quantities.
- Buy at most economical rates (quotations).
- Follow up operations. (Any 3 × 1) (3)

4.2
- Reject and return the damaged materials.
- Indicate the correction and ask the driver to sign for it on both copies of the deliver note. (Any 1 × 1) (1)

4.3
- Proper storage to avoid deterioration
- Proper control and check at gate and vehicles
- Proper lighting on the site at night
- Site security
- Must have a storage with lock and key
- A competent storekeeper
- Proper issuing of material requisitions
- Proper record-keeping of materials (Any 5 × 1) (5)

[9]

QUESTION 5

5.1
- Attendance register
- Minutes of the previous meeting
- Progress of work and causes of delay
- Labour, plant and material shortage
- Outstanding information
- Review of subcontractors
- General variation orders
- AOB and date for next meeting (6)

5.2
- Leadership
- Technical skill and knowledge
- Experience in the industry
- Punctuality and efficiency
- Capacity to allocate workers and subcontractors
- Take responsibility
- Loyal to the company (Any 5 × 1) (5)

5.3
- The need to belong
- The need for approval
- The need for status
- The need to achieve success
- The need for self-respect (Any 3 × 1) (3)

Copyright reserved

5.4 5.4.1 Practical completion is a list of defects and a date to fix them, given by the architect to the contractor.

5.4.2 Usually after 3 months of practical completion, the architect inspects for any defects again. If there are any, a list is given to the contractor to rectify and do the final completion.

5.4.3 These are sums which cannot be measured at the time BOQ is prepared, but enter in BOQ as provisional sums and later measured.

5.4.4 Any change in the original plan, will be notified by the architect to the contractor by issuing a variation order.

5.4.5 It is a 3-month period to the practical completion date.

5.4.6 Instead of going to court, any dispute between the contractor and the client is settled in front of an arbitrator who is usually mentioned in the contract.

(6 × 2) (12)
[26]

QUESTION 6

6.1
- Decide exactly what has to be delegated.
- The person must have the capacity the carry out the delegation.
- The person must fully understand what is expected of him/her.
- Ensure that others whom the person comes into contact with know about this delegation.
- Check occasionally that the tasks are being done properly. (Any 3 × 1) (3)

6.2
- Sound workmanship
- Work constructed at a fast rate of progress
- Work constructed at low cost (3)

6.3 Period planning
A specific period is taken out from the master plan for closer examination. (1)
Stage planning
A stage or a section of the contract is taken out for closer examination. (1)
[8]

QUESTION 7

Middle ordinates

$\frac{0 + 200}{2}$ $\frac{200 + 300}{2}$ $\frac{300 + 500}{2}$ $\frac{500 + 600}{2}$ $\frac{600 + 500}{2}$ $\frac{500 + 300}{2}$ $\frac{300 + 100}{2}$ (1)

$\frac{100 + 0}{2}$ $\frac{0 + 100}{2}$ $\frac{100 + 400}{2}$ $\frac{400 + 100}{2}$

100 250 400 550 550 400 200 50 50 250 250 (1)

Average height for cut

$\frac{100 + 250 + 400 + 550 + 550 + 400 + 200 + 50}{8} = 312,5$ (2)

Volume of cut (m) = height × distance × length
= 0,313 m × 1,95 m × 15 m
= 9,16 m³ (2)

Average depth for fill

$\frac{50 + 250 + 250}{3} = 275$ (2)

Volume of cut (m) = depth × distance × length
= 0,28 m × 0,75 m × 15 m
= 3,15 m³ (2)

Volume to be removed = cut - fill
= 9,16 – 3,15 = 6 m³ (2)
[12]

TOTAL: 100

Copyright reserved